

Chapter: 28

PARTICLE PHYSICS

DARK MATTER AND DARK ENERGY:

In particle physics, Dark Matter and Dark Energy are mysterious entities that make up about 95% of the universe's mass-energy budget.

1. DARK MATTER:

Definition:

Dark matter is a type of matter that does not emit, absorb, or reflect light, so it is invisible.

Explanation:

Dark Matter is thought to be composed of Weakly Interacting Massive Particles (WIMPs), such as axions or neutralinos.

Dark Matter interacts with normal matter only through gravity and the weak nuclear force, making them invisible to most detectors.

2. DARK ENERGY:

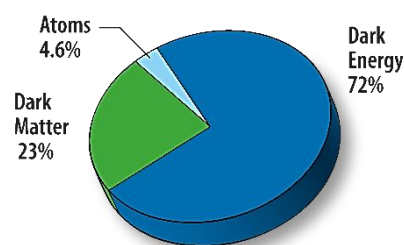
Definition:

Dark energy is an unknown form of energy that fills space and causes the universe to expand faster.

Explanation:

Dark Energy is believed to be a property of space itself, represented by a hypothetical field, like the Higgs field, that drives the accelerating expansion of the universe.

It means that it works like a **repulsive force**, pushing galaxies away from each other.



3. DARK SECTOR:

Definition:

The dark sector includes both dark matter and dark energy.

Explanation:

Together, they control the structure and evolution of the universe. They cannot be directly seen or detected. Scientists are still researching them, and many questions remain unanswered.

PARTICLE PHYSICS:

Definition:

Particle Physics is the branch of modern physics that studies the basic building blocks of matter (elementary particles) and the forces acting between them.

ELEMENTARY PARTICLES:

Definition:

Elementary particles are like tiny building blocks that combine to form everything in the universe.

Explanation:

People have always been curious about what the universe is made of.

- Earlier, scientists thought that **atoms** were the smallest units of matter.
- In the 1800s, it was discovered that atoms are made of **electrons, protons, and neutrons**.
- Today, we know there are even smaller particles called **elementary or fundamental particles**, which have no further parts.
- **Molecules, atoms, protons, and neutrons** are not fundamental because they are made of smaller particles.
- **Electrons, quarks, and photons** are fundamental particles because they cannot be divided further.

The main aim of particle physics is to study these particles and understand how they behave and interact.

STANDARD MODEL:

The universe is made of **matter and energy (radiation)**. To understand them, scientists divide particles into two main groups:

1. Matter particles
2. Force particles.

All these particles are explained in a theory called the **Standard Model of Particle Physics**.

THE STANDARD MODEL OF PARTICLE PHYSICS

Definition:

The Standard Model is a theory that explains three fundamental forces (electromagnetism, weak nuclear force, and strong nuclear force) and all known elementary particles.

Explanation:

The Standard Model is a **very successful theory**. It has predicted many new particles, and experiments have confirmed these predictions. However, it is **not complete**:

- It does not explain **gravity**.
- It also does not explain **dark matter and dark energy**.

CLASSIFICATION OF ELEMENTARY PARTICLES:

The Standard Model divides particles into two main types: (1) Fermions (2) Bosons

1. Fermions:

- These are matter-particles.
- They make up everything in the universe.

Simple Idea:

- Fermions → build matter
- Bosons → carry forces

2. Bosons:

- These are **force particles**.
- They carry forces and cause interactions between particles.

Remember: "Fermions build matter, Bosons carry forces."

FUNDAMENTAL FORCES AND THEIR FIELD PARTICLES:

ELEMENTARY OR FUNDAMENTAL FORCES:

Definition:

The smallest building blocks of matter are called elementary particles. But these particles alone cannot form the universe.

Example:

- Bricks make a house, but **cement is needed to join them**.
- In the same way, particles need a “**glue**” to hold them together.
- This “glue” is called **fundamental forces**.
- These forces help in **building and holding the universe together**.
- In the Standard Model, **each force has a special particle** connected with it.

FIELD PARTICLE:**Definition:**

A field particle is a virtual particle that transfers (carries) a fundamental force between two particles.

Explanation:

In the Standard Model, all field particles are placed in a group called **Bosons**.

Field particles are those particles which are connected with forces (force fields).

According to the Standard Model, **bosons are called field particles** because they are related to these forces.

Example:

The photon is a field particle associated with the electromagnetic field.

TYPES OF FUNDAMENTAL FORCES:

Scientists divide all fundamental forces into **four types** (from weakest to strongest).

1. Gravitational Force
2. Weak Nuclear Force
3. Electromagnetic Force
4. Strong Nuclear Force

1. GRAVITATIONAL FORCE:

It is the force that attracts two masses towards each other.

It is the weakest of the four fundamental forces but acts over long distances.

Field Particles: Gravitons

The hypothetical particle associated with gravity is graviton although it has not been observed yet.

The Standard Model doesn't explain gravity.

2. WEAK NUCLEAR FORCE:

It is the force responsible for radioactive decay.

It helps unstable nuclei **break into smaller stable nuclei**. It's also responsible for processes like beta decay and neutrino emission.

This force is weaker than electromagnetic and strong nuclear forces but stronger than gravitational force.

Field Particles: W^+ , W^- and Z Bosons

There are three field particles associated with weak nuclear force. These are w^+ boson, w^- boson and Z boson.

These short-lived bosons carry the force over very small distances, explaining the limited range of the weak force.

3. ELECTROMAGNETIC FORCE:

It is the force between charged particles (like electrons and protons).

It includes both electric and magnetic forces. The electromagnetic force is stronger than both gravitational and weak nuclear forces.

It is also a long-range force, similar to the gravitational force.

Field Particles: Photons

The field particles of electromagnetic force are photon. It is massless and charge-less particles.

Whenever charged particles interact, they exchange photons, causing the attractive or repulsive forces we observe.

4. STRONG NUCLEAR FORCE:

It is the force that holds the nucleus together.

The strong force binds quarks together to form protons and neutrons, and it holds atomic nuclei together.

It is the strongest force among all forces and acts at very small (subatomic) distances.

Field Particle: Gluons

Gluons are the field particles that mediate the strong force between quarks.

SUMMARY:

Force	Strength	Range	Field Particle
Gravitational	Weakest	Long	Graviton (not proven)
Weak Nuclear	Weak	Very Short	W^+ , W^- , Z
Electromagnetic	Strong	Long	Photon
Strong Nuclear	Strongest	Very Short	Gluon

PARTICLE ZOO:**Definition:**

The arrangement of all elementary particles in the Standard Model is called the Particle Zoo.

Explanation:

Scientists study particles by **grouping them into categories**, they find **similarities and differences** between them.

MAIN GROUPS OF PARTICLES:

In the Standard Model, particles are divided into two main groups based on **spin**:

1. Bosons
2. Fermions.

1. BOSONS:

Bosons are force-carrying particles with integer spin (0, 1, 2).

CLASSIFICATION OF BOSONS:

All bosons can be classified into two types:

1. Elementary Boson
2. Composite Boson.

1. ELEMENTARY BOSONS:

Elementary bosons are not made of smaller particles (elementary particles) and they carry fundamental forces.

Examples:

1. Gluon
2. W^+ , W^- and Z^0 Bosons
3. Photon
4. Higgs Boson (gives mass to all particles).
5. Graviton

The properties of elementary bosons:

Force	Boson	Spin	Strength	Mass
Strong	Gluon	1	1	Massless
Electromagnetic	Photon	1	10^{-2}	Massless
Weak	W, Z	1	10^{-7}	80.91 GeV
Gravity	Gravitation	2	10^{-39}	Massless

2. COMPOSITE BOSONS:

Composite bosons are made of the combination of quark and antiquark (even number of fermions).

These are called mesons and help in strong force between nucleons (protons and neutrons).

Examples:

Mesons are composite bosons that are composed of one quark and one antiquark.

2. FERMIONS:

Fermions are matter particles with half-integer spin ($1/2, 3/2$, etc.).

All matters in the universe are made of **fermions**.

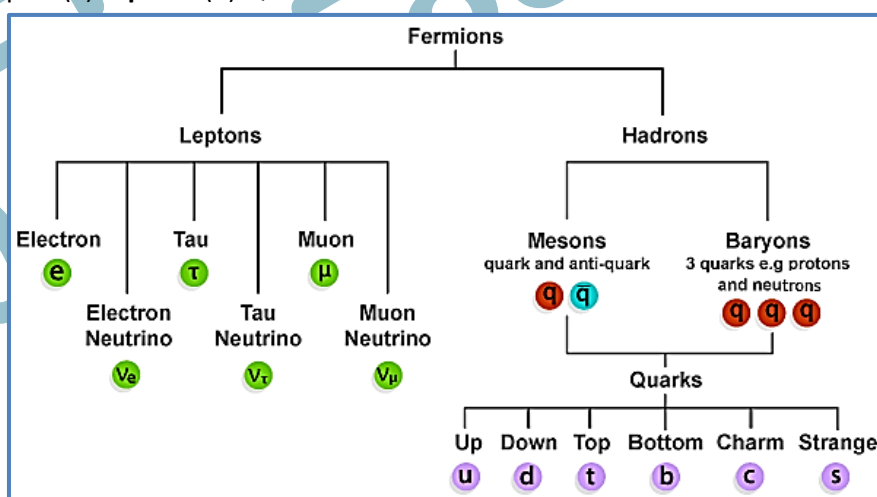
CLASSIFICATION OF FERMIONS:

All Fermions can be classified into two types: (1) Elementary Fermions (2) Composite Fermions.

1. ELEMENTARY FERMIONS:

Elementary fermions are basic building blocks of matter and are not made of small particles.

They are of two types: (a) **Leptons** (b) **Quarks**

**1(a) Leptons:**

Leptons are elementary particles that do not experience strong nuclear force.

Types of Leptons:

There are six types of leptons, which come in three pairs. The pairs are made up of three charged particles named electron (e^-), muon (μ), and tau (τ), along with their partners charge less neutrinos (ν).

Generations of Leptons and Quarks:

These all six leptons group into three generations:

- Generation: I = Electron + Electron neutrino
- Generation: II = Muon + Muon neutrino
- Generation: III = Tau + Tau neutrino

Each generation consists of one pair of leptons (for example, electron with electron neutrino).

	Fermions			
	Charge	Generation I	Generation II	Generation III
Leptons	-1	electron (e)	muon (μ)	tau (τ)
	0	e-neutrino	μ -neutrino	τ -neutrino
Quarks	+2/3	up (u)	charm(c)	top(t)
	-1/3	down (d)	strange (s)	bottom(b)

Properties of Leptons:

- Leptons interact via weak and electromagnetic forces but not through the strong force.
- Leptons are stable particles and do not undergo decay under normal circumstances.
- Leptons exist alone and do not form groups.

1(b) Quarks:

Quarks are elementary particles that experience all three fundamental forces (strong nuclear force, weak nuclear force, and electromagnetic force).

Quarks are a fundamental component of visible matter. All the matter around us, such as protons and neutrons, is made up of quarks. It means that:

- Quarks make **protons and neutrons**
- So, all visible matter is made of **quarks**

Types of Quarks:

Quarks come in six types, or flavors: up (u), down (d), charm (c), strange (s), top (t), and bottom (b).

Generations of Quarks:

Quarks also have **3 generations**, with each generation containing one pair of quarks.

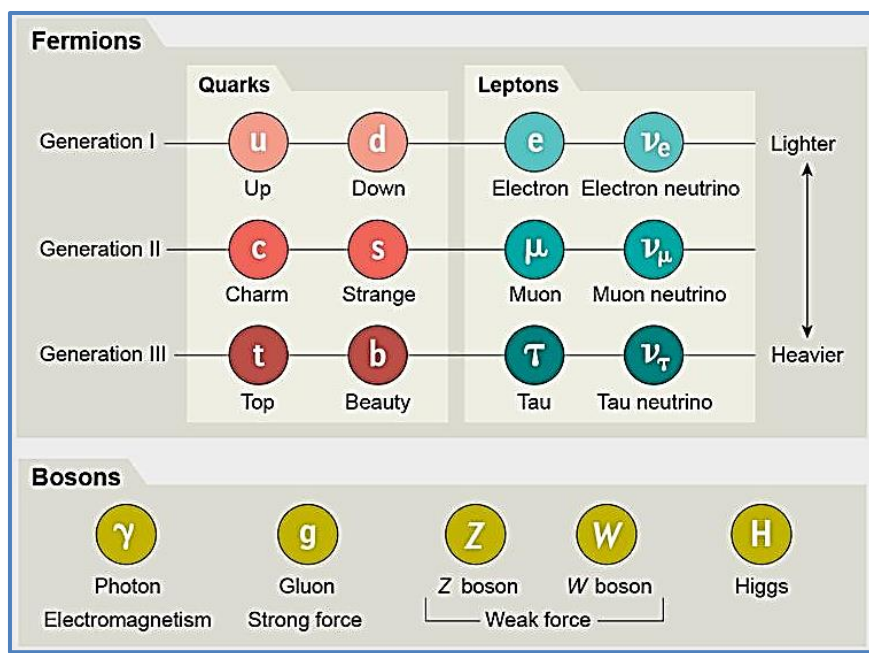
The up and down quarks are the only stable quarks in ordinary matter.

Properties of Quarks:

- They carry fractional electric charges, either +2/3 of e or -1/3 of e.
- They interact through strong force (via gluons).
- Quarks are never found as free particles in nature; they are always confined within larger particles called hadrons.
- Hadrons are particles made up of quarks held together by the strong nuclear force.
Examples of hadrons include protons and neutrons (Proton and neutron are **hadrons**).
- Quarks can change type by **weak interaction** (example: beta decay)
- Weak interactions can change one type of quark into another.
For example, a down quark can change into an up quark through weak decay processes.

SELF-ASSESSMENT QUESTIONS:

1. Which particles don't participate in strong nuclear force?
2. Define the term "flavor" in the context of quarks and list the six quark flavors.

**TYPES OF CHARGE:**

In physics, there are two types of charges: 1. Electric charge 2. Color charge (strong charge)

1. ELECTRIC CHARGE:

Electric charges have two types: (1) Positive (+) (2) Negative (-)

2. COLOR CHARGE:

Color charge is a type of charge of quarks related to the strong nuclear force.

Explanation:

Color charge is similar to electric charge, but instead of just **being positive and negative**, it has **three types**:

1. Red
2. Green
3. Blue

These are only **names (labels)**, not real colors. Quarks interact with each other through strong **force**.

Characteristics:

1. Quarks carry **one color charge** (red, green, or blue) and can **change their color** during interactions.
 - Quarks of **different colors attract each other**.
 - Quarks of the **same color repel each other**.
2. Quarks always combine in such a way that their colors cancel and form a **neutral (white) particle**.
For example: a proton consists of three quarks: one red, one green, and one blue, making it color neutral.

A proton = red + green + blue $\rightarrow$ color neutral (white)

3. Free quarks **cannot exist alone**. They are always found in groups. This is called **quark confinement**.

2. COMPOSITE FERMIONS:

Composite subatomic particles (such as protons, neutrons, alpha particles etc.), are composed of two or more elementary particles.

All composite particles are massive.

Examples: proton, neutron, alpha particle

HADRONS:

The composite particles that are made of quarks are called hadrons.

The two main categories of hadrons are: (i) Baryons (ii) Mesons.

2(i) Baryons:

Baryons are a class of hadrons that consist of three quarks.

For example, protons and neutrons are the most well-known baryons, each composed of three quarks:

Proton:

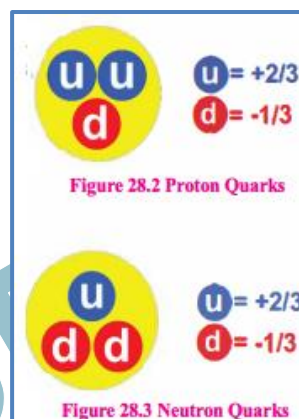
It consists of two up and one down quarks (uud). Up (u) quark has $+2/3$ charge and down quark has $-1/3$ charge. Therefore, net charge on proton has $+1$ charge:

$$\text{Proton} = uud = \frac{2}{3} + \frac{2}{3} - \frac{1}{3} = +1$$

Neutron:

It consists of three quarks: one up and two down quarks (udd). The net charge on neutron is zero:

$$\text{Neutron} = udd = \frac{2}{3} - \frac{1}{3} - \frac{1}{3} = 0$$



2(ii) Mesons:

Mesons are another class of hadrons, but they consist of one quark and one anti quark.

For example, pions (π) are common Mesons.

They carry strong nuclear force, binding protons and neutrons together.

QUARKS AND BETA DECAY:

Definition:

β -decay is a radioactive decay process in which a neutron changes into a proton or a proton change into a neutron in the atomic Nucleus.

1. β^- -DECAY:

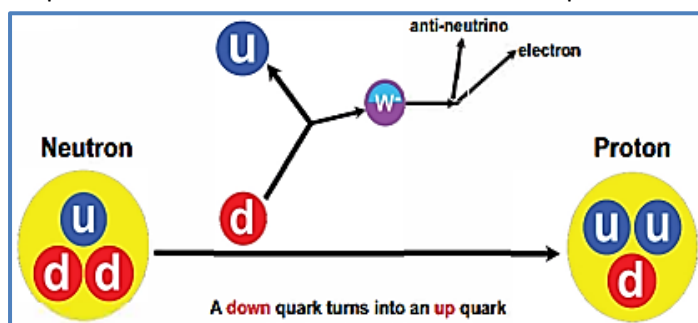
Definition:

β^- -decay is a type of radioactive decay process in which a neutron is transformed into a proton.

Explanation:

When a β^- -decay occurs, a down quark (d) emits a field particle called a W^- boson (a weak force carrying particle) and transforms into an up quark (u).

The W^- boson quickly breaks up into an electron and an antineutrino. The process is shown in Figure.



Electric charge is conserved during beta decay with the emission of an electron and an antineutrino.

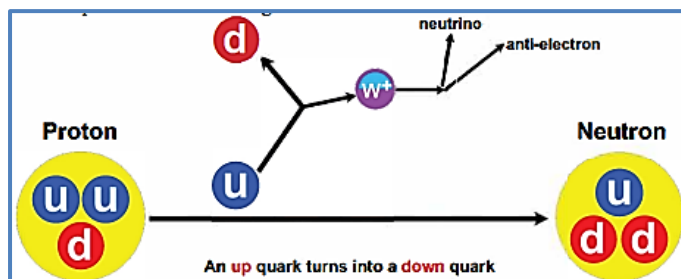
2. β^+ -DECAY:

Definition:

β^+ -decay is a type of radioactive decay process in which a proton is transformed into a neutron.

Explanation:

In β^+ -decay, an up quark (u) emits a field particle called a W^+ -boson and transforms into a down quark (d). The W^+ -boson quickly breaks up into an anti-electron (positron) and a neutrino. The process is shown in Figure.



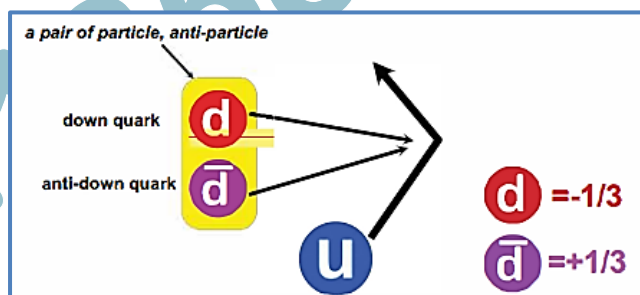
Q: The up quark can change into a down quark by emitting W^+ weak particles. How could that trick be done? Explain.

Ans: Nature can create short-lived particles, like particle-antiparticle pairs, out of nothing by borrowing energy permitted by the uncertainty principle.

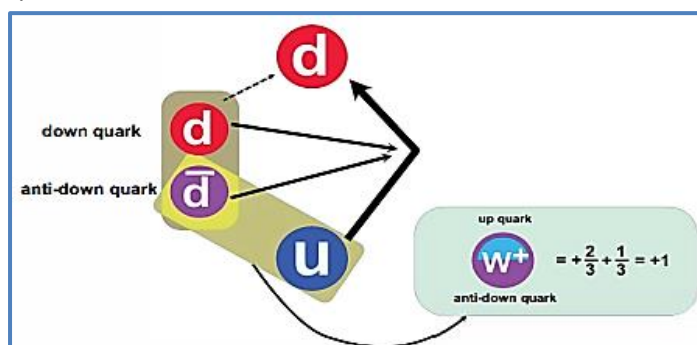
This principle allows for a fluctuation in energy levels at any given moment, making it possible for particles to briefly come into existence before annihilating each other.

Imagine that there was created such a pair of particles: a down quark and anti-down quark in the neighborhood of the up quark:

Anti-down quark is represented by bar over symbol d ($\bar{d}$). The charge on anti-down is $+1/3$. Therefore, net charge on the pair is zero.



Next, imagine the down quark replacing the up quark. The up quark could then join up with the remaining anti-down quark to make the W^+ particle:



RADIATIONS DETECTORS:

Anti-down quark is represented by bar over symbol d: $\bar{d}$. The charge on anti-down is $+1/3$. Therefore, net charge on the pair is zero.

Definition:

Radiation detectors are devices used to detect, track, and identify ionizing particles produced from nuclear decay, cosmic rays, or particle experiments.

Explanation:

These detectors not only tell us that a particle is present, but they can also measure its energy, charge, momentum, spin, and even identify the type of particle.

Many detectors work on a simple idea: when a charged particle passes through a material, it **ionizes atoms** (removes electrons). This ionization is then detected and measured.

Classification:

Particle detectors are classified into numerous types, including ionization detectors, scintillation detectors, Cherenkov light detectors, transition radiation detectors, and others.

GASEOUS DETECTORS:

Definition:

The detectors based on the loss of energy caused by the ionization of atoms are called gaseous detectors.

Explanation:

Some detectors use **gas** as a medium. When a charged particle passes through the gas, it causes **ionization** and loses energy.

This ionization is converted into an **electrical signal**, which helps in detecting the particle and finding its path. Because they show the **path (track) of particles**, they are also called **tracking detectors**.

Examples:

1. Wilson Cloud Chamber
2. Geiger-Muller (GM) Counter

1. WILSON CLOUD CHAMBERS:

Introduction:

The Wilson Cloud Chamber was invented by Scottish physicist Charles Thomson Rees Wilsonin (1912). It helped in the discovery of many **subatomic particles**. It is a tracking detector based on ionization.

Definition:

The Wilson Cloud Chamber is a device used to see the tracks of subatomic particles.

Explanation:

A Wilson cloud chamber is a type of tracking detector that works on the principle of ionization.

It is a class of gas detectors that are used in particle physics and nuclear physics to visualize the tracks of subatomic particles such as electrons, positrons, alpha particles, and cosmic rays.

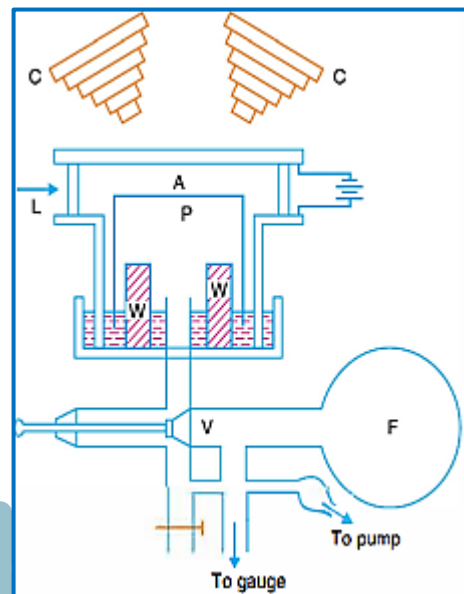
WORKING PRINCIPLE:

The Wilson Cloud Chamber consists of a sealed container filled with supersaturated vapor, typically water or ethanol.

When a particle passes through the chamber, it ionizes the vapor, creating a trail of droplets that condense around the ionized path. This creates a visible cloud-like track that can be photographed and analyzed.

CONSTRUCTION:

- The Wilson cloud chamber consists of a **large cylindrical glass chamber (A)** containing **dust-free air saturated with water vapor**.
- A **piston (P)** is fitted inside the chamber. This piston is connected to an **evacuated vessel (F)** through a **valve (V)**.
- When the valve is opened, the air below the piston moves into vessel **F**, causing the piston **P** to move down suddenly.
- **Wooden blocks (WW)** are used to reduce the air space under the piston. Water is present at the bottom to keep the air **fully saturated**.
- The amount of expansion can be controlled by changing the **height of piston (P)**.
- A strong **light source (L)** and **two cameras (CC)** are arranged to observe and record the tracks.



WORKING:

- When the valve **(V)** is opened, air rushes from chamber **A** into vessel **F**, and the piston **(P)** drops quickly. This causes **sudden (adiabatic) expansion** of air in chamber **A**, making it **cool and supersaturate**.
- At this moment, ionizing particles enter chamber **A** through a side window. These particles **ionize the air**, producing ions along their path.
- Very fine **water droplets form around these ions**, creating a **visible track** of the particle.
- The chamber is then illuminated with a strong **light (L)**, and **cameras (CC)** take photographs of these tracks.
- All these steps must happen **very quickly** to get clear tracks.

TRACKS OF PARTICLES:

Different particles produce different tracks:

- **Alpha particles:** Produce thick, straight tracks
- **Beta particles:** Produce thin, curved tracks

CONCLUSION:

The cloud chamber has helped in discovering particles like **positron and meson**.

USE OF WILSON CLOUD CHAMBERS:

Wilson cloud chambers are very useful in studying particles and their behavior.

1. Particle Identification:

Scientists use the cloud chamber to **identify different particles**. By observing the **shape and curvature of tracks**, they can recognize and classify particles.

2. Nuclear Physics Research:

Cloud chambers have been used to study the behavior of particles in nuclear reactions and to investigate the structure of atomic nuclei.

3. Cosmic Ray Studies:

The cloud chamber helps in detecting **cosmic rays**, which are **high-energy particles coming from space**, and show their paths.

4. Education and Outreach:

Cloud chambers are often used as educational tools in physics classrooms and science museums to help students and the general public visualize the behavior of subatomic particles.

2. GEIGER-MULLER COUNTER:

Introduction:

The Geiger–Muller counter was invented by Hans Geiger and Walther Muller (1928). It is a gas radiation detector used to measure ionizing radiation.

Definition:

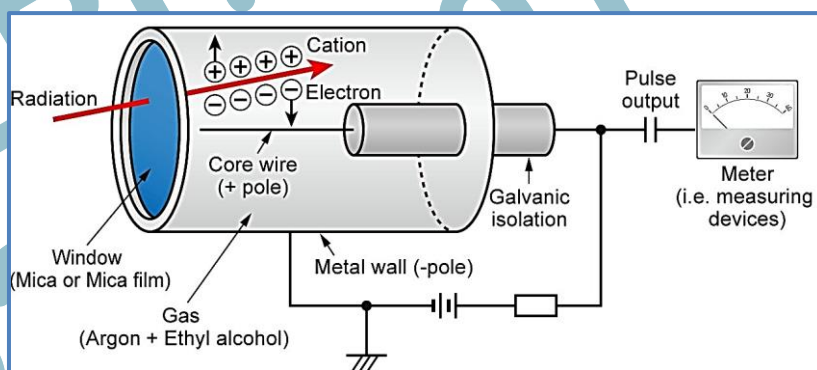
A Geiger-Muller counter (GM counter) is a type of gas radiation detector used to measure ionizing radiation.

Explanation:

It is a portable and versatile device commonly employed for detecting the presence and intensity of various types of ionizing radiation, including alpha particles, beta particles, and gamma rays.

CONSTRUCTION:

- The GM counter consists of a **metallic cylindrical chamber** which acts as the **cathode**.
- A **thin wire anode** is placed along its axis.
- The chamber has a **thin window** through which radiation enters.
- The chamber is filled with inert gas (helium, neon, or argon) at low pressure.
- A **high voltage supply** is connected across the tube.
- A **counter** is attached to record the radiation.



WORKING:

- When radiation enters the chamber through the window, it **ionizes the gas** inside. This produces electrons and ions.
- Due to the applied high voltage, the electrons move quickly towards the **anode** and cause **more ionization (chain reaction)**.
- This produces a large number of electrons which reach the anode and create a **pulse of current**.
- The counter records these pulses, and in this way the **amount of radiation is measured**.

RESULTANT CURVE: GEIGER PLATEAU

The working of GM counter is shown by a graph between **count rate and voltage**.

1. Initial Region:

At low voltages, there is **almost no ionization**, so the count rate is very low.

2. Threshold Region:

As voltage increases ionization starts, the counter starts to detect ionizing events. The count rate begins to rise sharply.

3. Geiger Plateau:

At a certain voltage range, the count becomes **constant**. This is called the **Geiger Plateau** (best working region).

The plateau indicates that each ionizing event creates a detectable pulse, and the counter is functioning reliably.

4. Continuous Discharge Region:

If voltage is increased too much, **continuous discharge** occurs and the counter does not work properly.

USE OF GM COUNTER:

The GM counter is widely used to detect radiation:

1. Radiation Monitoring:

It is used in **nuclear plants, labs, and industries** to measure radiation and ensure safety.

2. Environmental Monitoring:

It is used for **environmental monitoring** to check background radiation.

3. Health Physics:

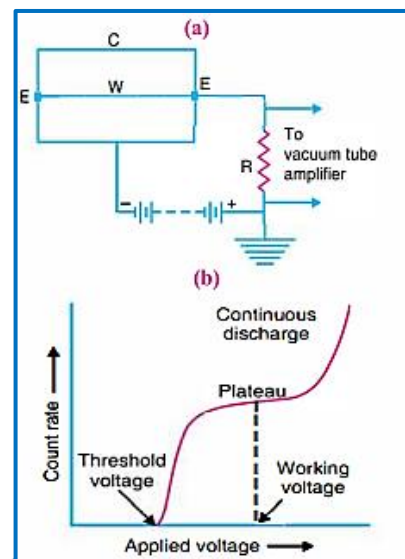
It helps in **health physics** to measure radiation exposure of workers.

4. Education:

GM counters are used in schools and colleges to demonstrate and study radiation and its effects.

5. Radiological Emergencies:

It is used in **radiological emergencies** to detect radiation leakage or contamination.



SUMMARY

- Gravitational Force is weakest, acting on mass, mediated by the hypothetical graviton.
- Electromagnetic is a force that acts on electric charge, mediated by the photon.
- Strong Nuclear Force is the strongest force; it acts on quarks and gluons within hadrons mediated by gluons.
- Weak Nuclear Force is responsible for radioactive decay, mediated by W and Z bosons.
- Standard Model of Matter Hadrons is composite particles made up of quarks, including baryons (e.g., protons and neutrons) and mesons.
- Leptons are fundamental particles that are not made of quarks, including electrons, muons, taus, and their associated neutrinos.
- Quarks are fundamental particles that combine to form hadrons, come in six flavors (up, down, charm, strange, top, bottom).
- Radiation Detectors are devices used to detect and measure ionizing radiation for safety, environmental monitoring, and scientific research.
- Wilson Cloud Chambers

Working Principle: It visualizes paths of charged particles through supersaturated vapor, forming condensation trails.

Construction: Sealed chamber with alcohol vapor and cooled base.

Use: Detecting and studying the tracks of ionizing particles, observing particle interactions.

- Geiger-Muller (GM) Counter

Working Principle: Detects ionizing radiation by ionizing gas within a tube, generating electrical pulses.

Construction: Tube filled with inert gas, central wire anode, outer cathode, high voltage applied.

Use: Measuring radiation levels, monitoring radioactive contamination, ensuring safety in nuclear facilities and laboratories.

TEXTBOOK EXERCISE:

SECTION- A: MULTIPLE CHOICE QUESTIONS (MCQS)

1. The Standard Model classifies elementary particles into two main groups:
(a) Baryons and Leptons (b) Fermions and Bosons (c) Quarks and Gluons (d) Hadrons and Mesons
2. The following is NOT a flavor of quark:
(a) Up (b) Down (c) Electron (d) Top
3. The charge of an up quark is:
(a) $+1/2e$ (b) $-1/2e$ (c) $+2/3e$ (d) $-2/3e$
4. The concept of "quark confinement" implies that:
(a) Quarks cannot exist as free particles outside of hadrons.
(b) Quarks are always found in pairs with opposite charges.
(c) Quarks have a strong affinity for gluons.
(d) Quarks are the fundamental building blocks of all matters.
5. The primary role of the Higgs boson in the Standard Model is:
(a) Mediating electromagnetic interactions (b) Providing mass to other particles
(c) Transmitting the strong nuclear force (d) Creating dark matter
6. According to the Standard Model, the term "color" refers to:
(a) Visible light spectrum (b) Charge property of quarks

- (c) Mass of particles (d) Spin of particles
7. A particle made up of a quark and an antiquark is called:
(a) Lepton (b) Baryon (c) Meson (d) Neutrino
8. An elementary particle that feels all three fundamental forces (electromagnetic, weak, and strong nuclear forces) is:
(a) Lepton (b) Quark (c) Electron (d) Neutrino
9. The primary function of a Geiger-Muller counter in particle physics is to:
(a) Measure the velocity of particles (b) Detect and count ionizing radiation
(c) Create antimatter particles (d) Generate magnetic fields
10. In a Geiger-Muller counter, the gas commonly used to detect ionizing radiation is:
(a) Oxygen (b) Neon (c) Argon (d) Helium

SECTION- B: SHORT ANSWERED QUESTIONS

1. Explain the difference between bosons and fermions and their roles in mediating fundamental forces.
2. Compare and contrast the properties of quarks and leptons, the two main categories of fermions in the Standard Model.
3. Define the term "lepton" and provide examples of leptons. Explain their fundamental properties and role in the Standard Model of particle physics.
4. Explain the concept of color charge in quarks and its significance in the strong nuclear force. How does the combination of quarks contribute to the color-neutral nature of protons and neutrons?
5. Describe the structure of a proton and neutron in terms of its quark composition. How do quarks combine to form a proton and a neutron, and what are the specific types of quarks involved?

SECTION- C: LONG ANSWERED QUESTIONS

1. Explain the structure of the Standard Model, including the different types of particles and their relationships. How does the model classify fundamental forces?
2. Describe in detail all fundamental forces and their associated field particles. What is boson, and why are bosons referred to as field particles?
3. Describe the operating principle of a Geiger-Muller counter. How does it detect and quantify ionizing radiation, and what are its limitations in terms of measurement range and types of radiation detected?
4. Discuss the uses of a Wilson cloud chamber in particle physics experiments. How does it visualize charged particle tracks, and what factors can affect its performance?